BINHUA HUANG

Curriculum Vitae

Center for Intelligent Bionic
Shenzhen Institute of Advanced Technology
(+86) 17710114101
bh.huang@siat.ac.cn
My Webpage
Github

Education

April,2022 **M.S. Computer Science**, *University of Science Malaysia*, GPA: 3.47/4.
Courses: Research and Empirical Methods in Computer Science, Advanced Algorithms & Complexity, Machine Learning, Wireless and Mobile Communications, Internet of Things
Dissertation: Tactile Sensing Using Machine Learning Methods (109 pages, in English)

July,2015 **B.E. Electronic Information Engineering**, *Longyan University*, GPA: 79.33/100, top15%.
Courses: C Language, Object Oriented Programming, MATLAB Programming, Embedded Systems and Applications, VHDL Language, Signals and Systems, Electronics, Digital Signal Processing, Automatic Control, Communication Principle, Sensors, Microcomputer etc.
Thesis: An IoT System Based on Cloud Computing (Best thesis award, 2%)

Publications

Published:

2022 **Modeling and control strategy of flexible joint servo system in humanoid manipulator driven by tendon-sheath**.
Meng Yin, Dongyang Shang, **Binhua Huang**, Xinyu Wu, and Wei Fen

2022 **RBF network-based adaptive sliding mode control strategy for the tendon-sheath driven joint of a prosthetic hand**.
Meng Yin, **Binhua Huang**, Zhengkun Yi, and Shibo Cai

Accepted:

2022 ***Methods to Recognize Depth of Hard Inclusions in Soft Tissue Using Ordinal Classification for Robotic Palpation***.
Zhenning Zhou, **Binhua Huang**, Runzhi Zhang, Meng Yin, Chengliang Liu, Yiwen Liu, Zhengkun Yi, and Xinyu Wu
The manuscript was accepted by *IEEE Transactions on Instrumentation & Measurement*.

2022 **Depth Recognition of Hard Inclusions in Tissue Phantoms for Robotic Palpation**.
Zhenning Zhou, Senlin Fang, Chaoxiang Ye, Tingting Mi, **Binhua Huang**, Xiaoyu Li, Zhengkun Yi, and Xinyu Wu
The manuscript was accepted by *IEEE International Conference on Real-time Computing and Robotics (RCAR 2022)* and as the best student paper.

2022 ***Multi-kernel-size Convolutional Supervised Autoencoders for Tactile Gesture Recognition***.
Chaoxiang Ye, Xiaoyu Li, **Binhua Huang**, Yuanzhe Su, Tingting Mi, Zhenning Zhou, Zhengkun Yi, and Xinyu Wu
The manuscript was accepted by *IEEE International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER 2022)*.

Under review:

2022 ***Liquid Volume Estimation by Tactile Sensor Using Machine Learning***.
Binhua Huang, Meng Yin, Zhengkun Yi, Chaoxiang Ye, Xiaoyu Li, Zhenning Zhou, and Xinyu Wu
The classification accuracy for the task of liquid volume estimation reached 99.78% with CNN; it is a considerable improvement compared to the state-of-the-art vision-based solution. The paper was submitted to *IEEE Transactions on Instrumentation and Measurement*.

2022 ***An IoT Based SCD Risk Rescue Alert System***.
Binhua Huang, Selvakumar Manickam, and Shafiq Ul Rehman
We proposed an IoT system that can detect the Sudden Cardiac Death (SCD) in real-time and ask for help automatically. The paper was submitted to *KSII Transactions on Internet and Information Systems*.

Manuscript in preparation:

2022 ***An Optical Wearable Device For ASL Recognition Using Neural Network***.
Binhua Huang, Chaoxiang Ye, and Zhengkun Yi
We designed a novel wearable device for American sign language (ASL) recognition, reaching 99.9% accuracy. It has the potential to be used in real-time ASL translation for deaf mutism films or TV shows. The experiment was completed. Now in the writing process.

2022 ***A Novel Tactile Sensor Mimics the Function of Ruffini Receptors Applied in Localization***.
Binhua Huang, Chaoxiang Ye, Meng Yin, Zhengkun Yi, and Xinyu Wu
We propose a new tactile sensor and use a neural network to localize touch areas and forces.

Research Experience

Shenzhen Institute of Advanced Technology, Chinese Academic of Science

August 2021 - Present ***Visiting Student (as well as RA)***.
Developing a deep multi-modal architecture for accurately predicting protein interaction information from biomedical text.

Advisor : **Dr. Zhengkun Yi**, *Professor, Center for Intelligent Bionic*, SIAT (*Personal Web-page*)

Skills

Programming Languages	Python, C, C++, VB
Frameworks and Platforms	Sci-kit, Pytorch, keras, Linux, Universal Robotics, MCUs, Arduino, Raspberry Pi, FPGA
Software and Tools	Matlab, LaTex, Altium Designer, Multisim, Proteus, Quartus II, Keil, IAR
Natural Languages	Chinese (mother tongue), English (CET-6, M.S. degree conducted in English)

Fellowships & Awards

2020	Second prize of Fujian provincial innovation competition
2015	Best graduation thesis award
2014	Third-class scholarship
2012	First prize of campus electronic competition
2012	Third-class scholarship
2011	Third-class scholarship

Referees

Dr. Zhengkun Yi
Professor
Center for Intelligent Bionic
Shenzhen Institute of Advanced Technology
✉ zk.yi@siat.ac.cn

Dr. Putra Sumari
Professor
School of Computer Sciences
University of Science Malaysia
☎ +(604) 653-3615
✉ putras@usm.my